

BAAZ JKBOSE

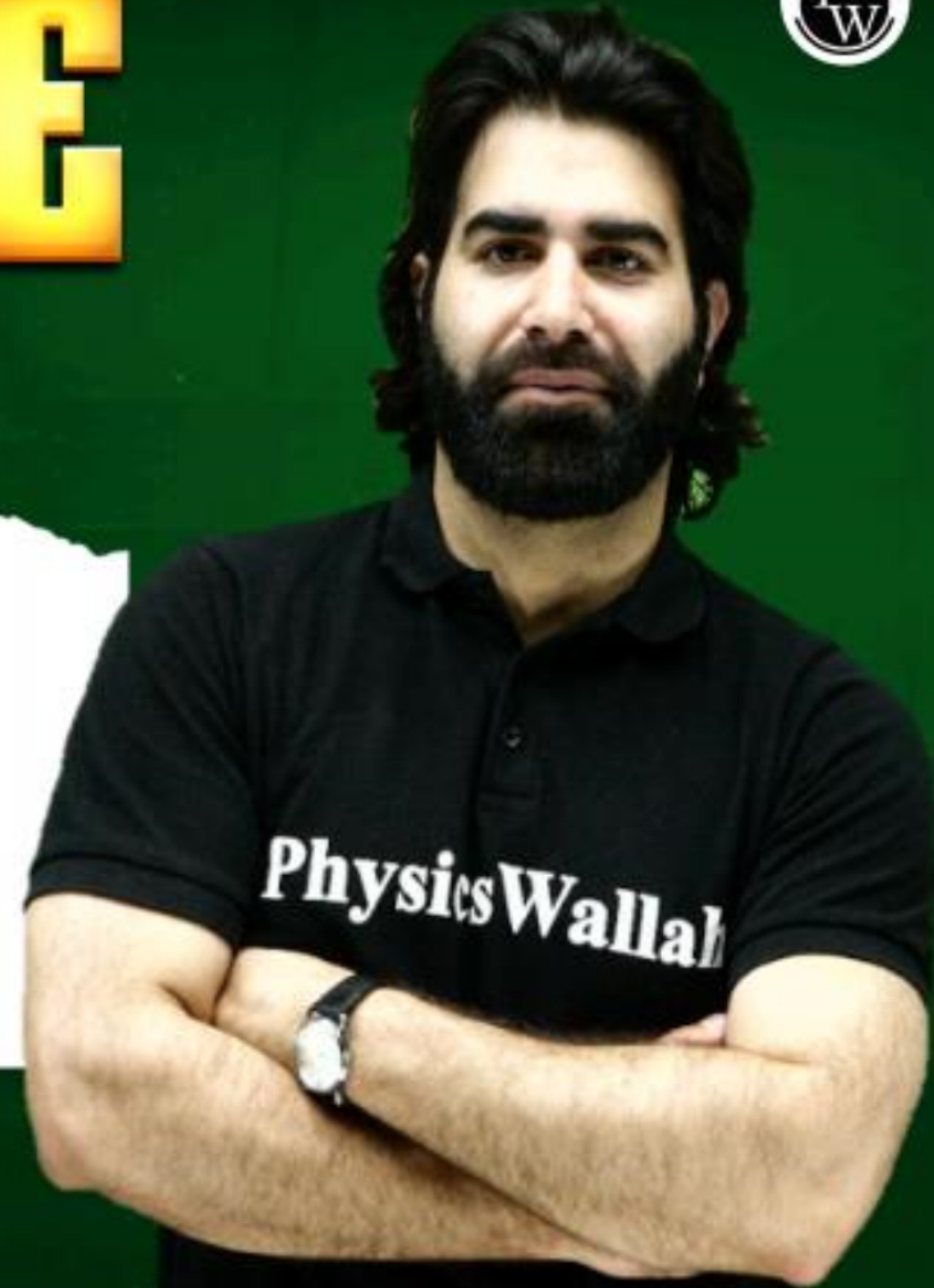
2026 CLASS-11TH

Body Fluids and Circulation

LECTURE-03

Zoology

Dr Adil Altaf sir

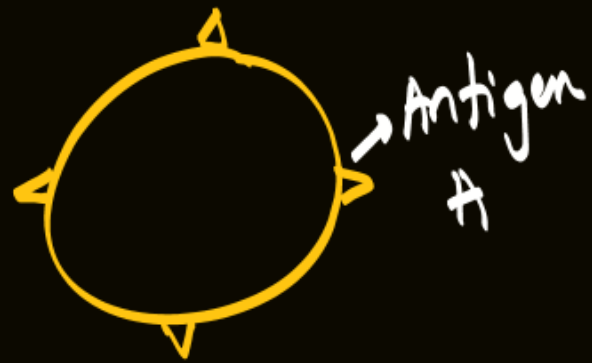


Topics To Be Covered

- 1 Erythroblastosis foetalis, clotting

Revision

ABO-type



1. A (Blood group)



2. O (Blood group)

<u>Blood</u>	<u>Antibody</u>
Antigen A	3. <u>B</u>
Antigen B	4. <u>A</u>
Antigen A, B	5. <u>No Antibody</u>
Antigen 'No'	6. <u>A, B Antibodies</u>

Continue - (ABO type)

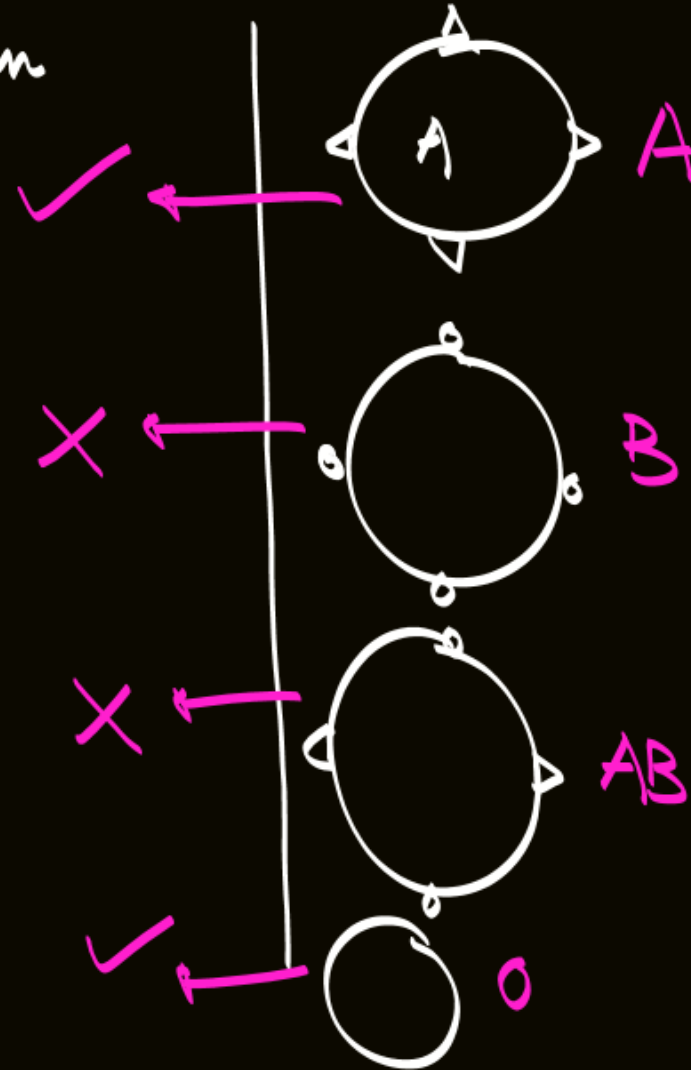
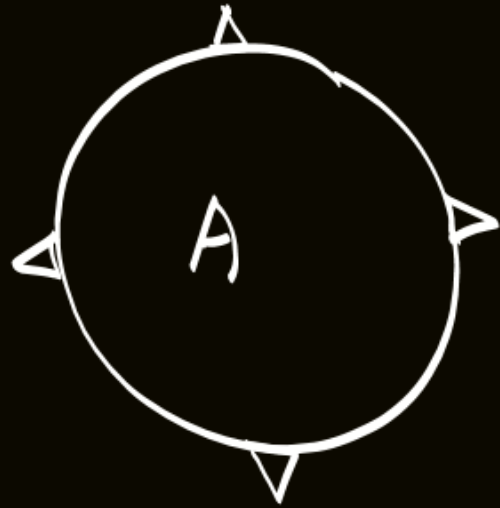
- Transfusion - Here matching done in a way that Donor Blood is not destroyed by Antibodies of Reeipient.

- Reeipient - Blood group A

→ Antigen = A

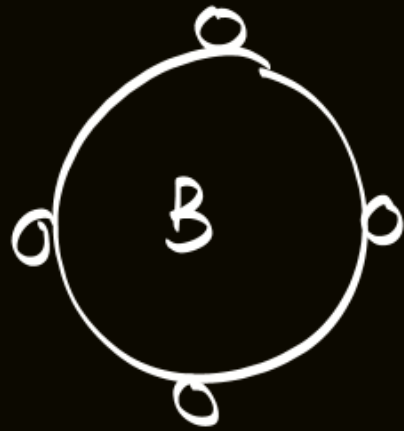
→ Antibody = B

↓
It will destroy B Antigen
Blood.

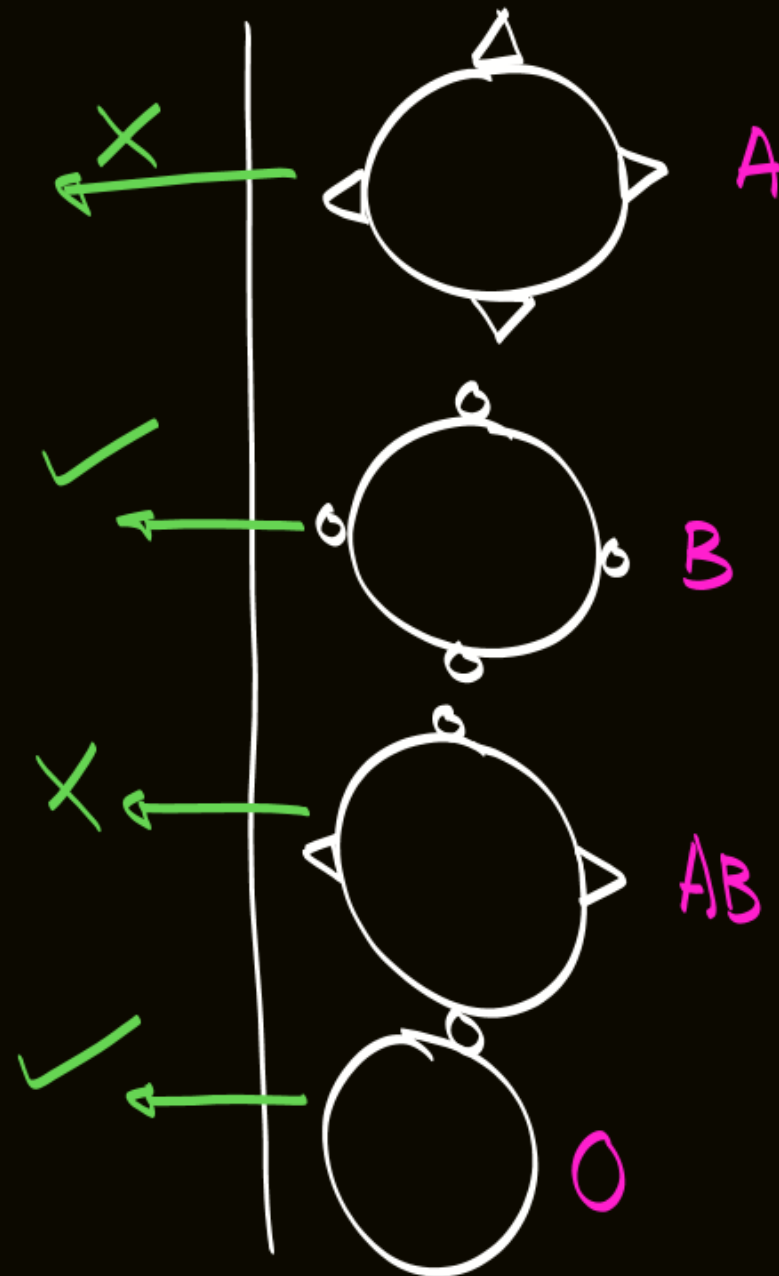


Receipient - Blood group B.

- Antigen B
- Antibody A



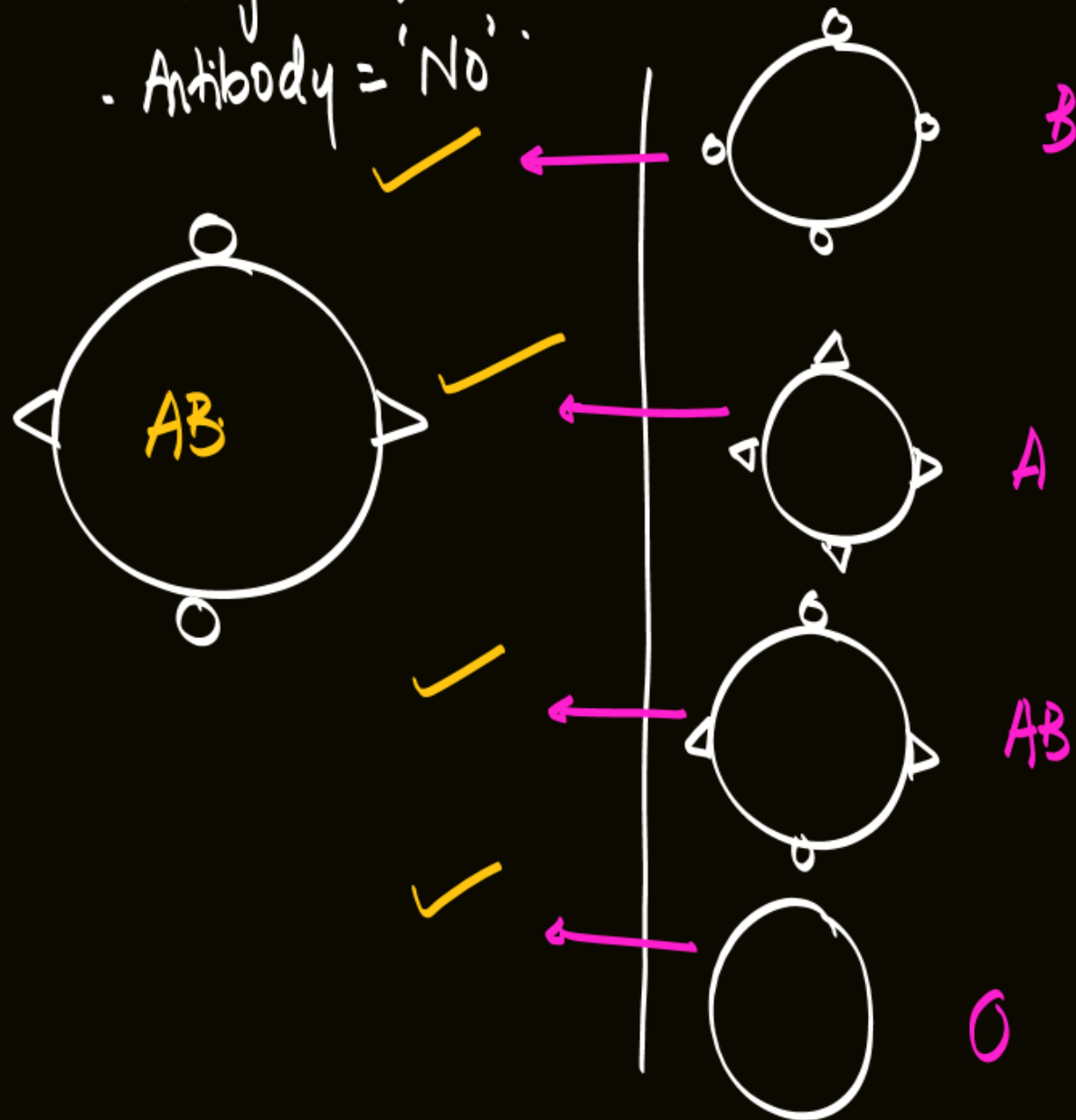
Donar B, O



Recipient Blood group AB.

Donor A, B, AB, O
(Universal Acceptor)

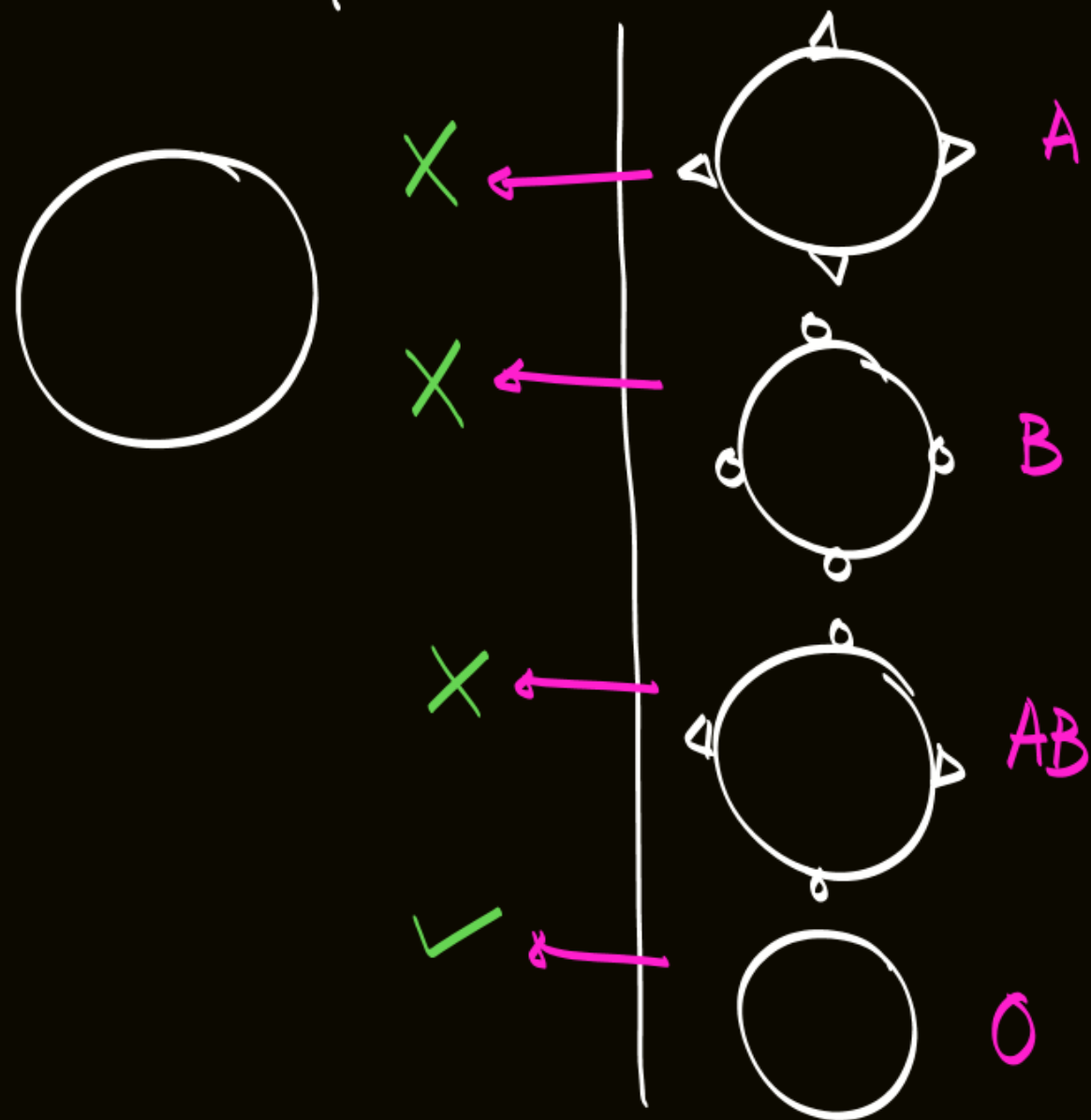
- Antigen A, B
- Antibody = 'No'.



Recipient Blood group O

- No Antigen
- Antibody = A, B

Donor O
Universal Donor O.



→ Rh type

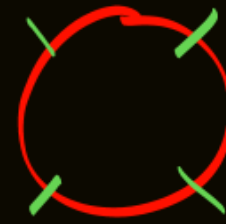
Rh is a Rhesus Blood group.

- Rh was discovered in Rhesus Monkey.

→ person, whose RBC have Rh Antigen (D-Antigen) is called Rh positive ($Rh^+ Rh^+ / Rh^+ Rh^-$)

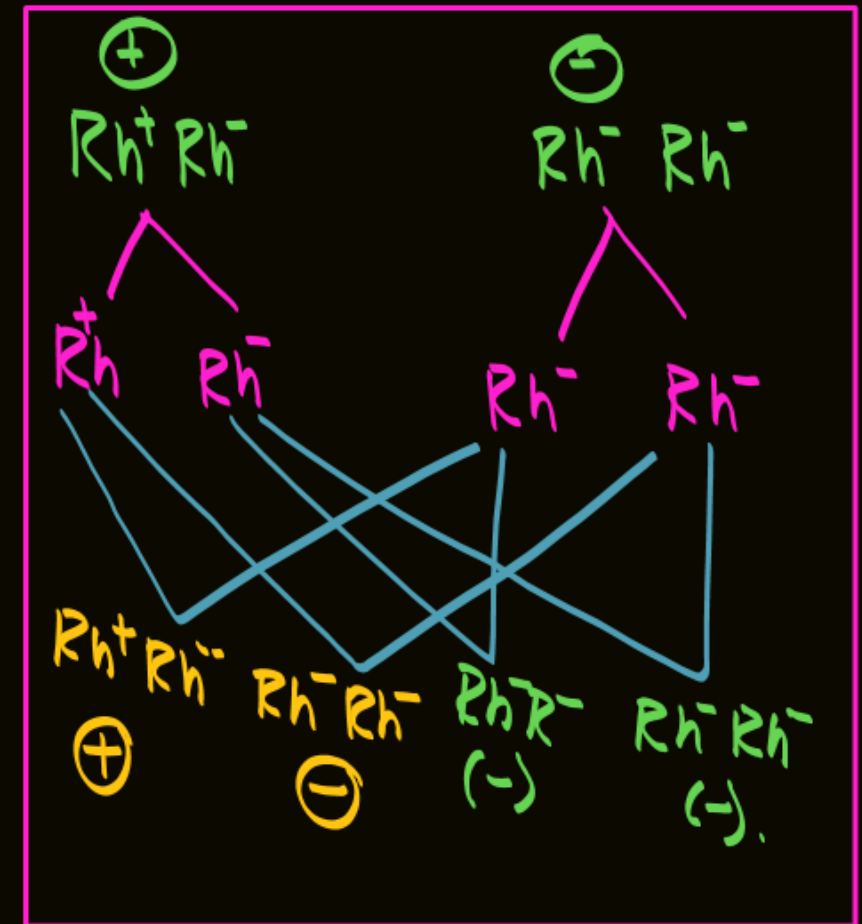
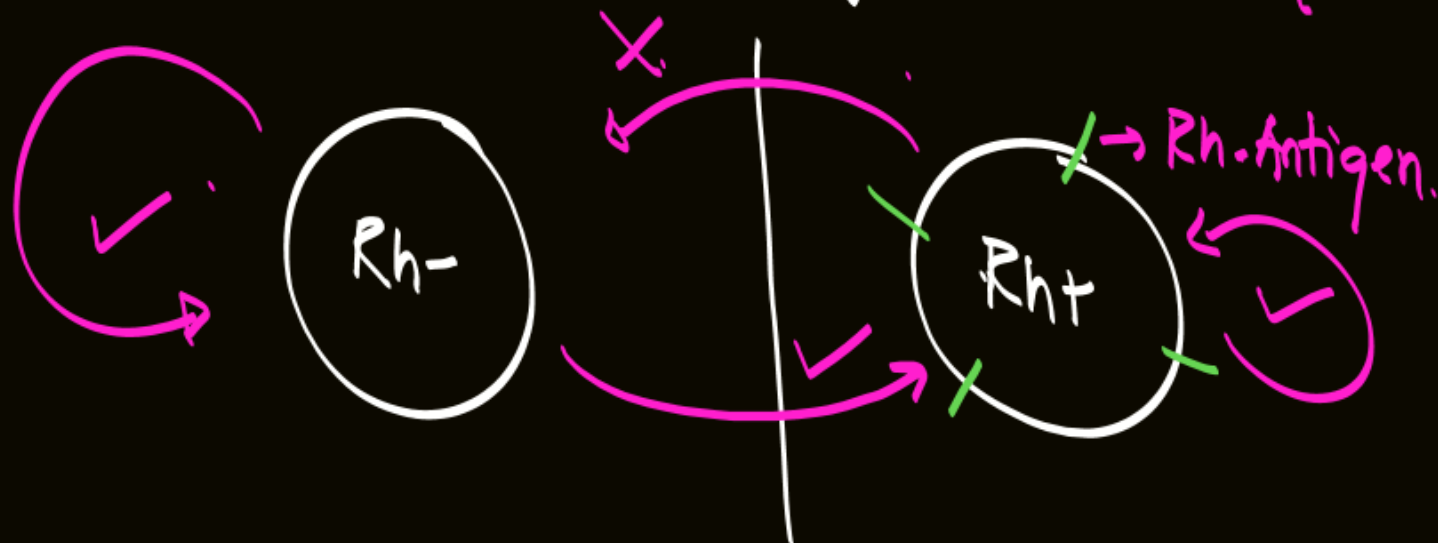
→ person, whose RBC don't have Rh Antigen, is called

Rh negative
($Rh^- Rh^-$).

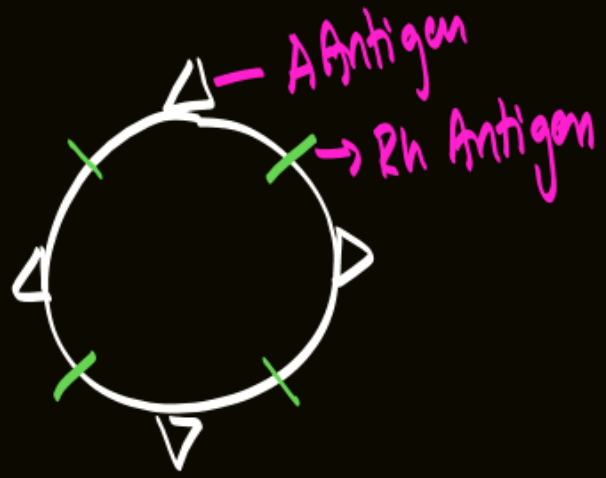


→ Rh Transfusion → Rh^- can give Blood to Rh^+

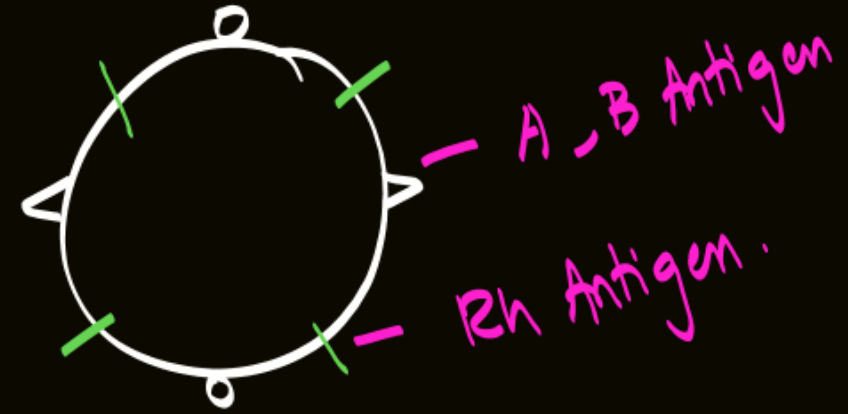
But Rh^+ can't give Blood Rh^- [Because Rh Antibodies will be produced by $Rh(-)$]



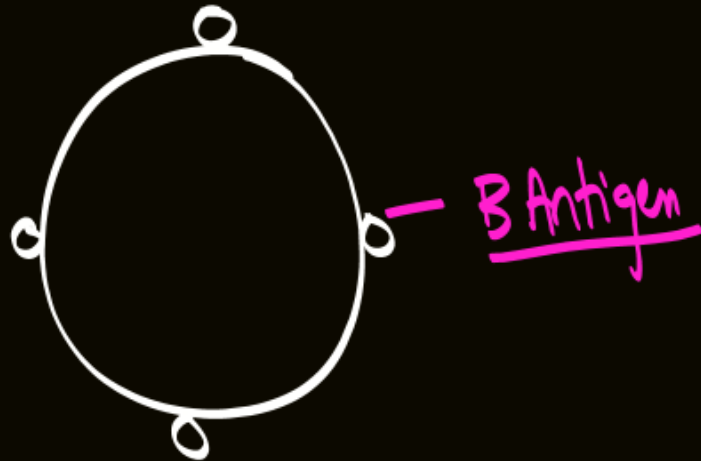
Now Combining Rh & ABO.



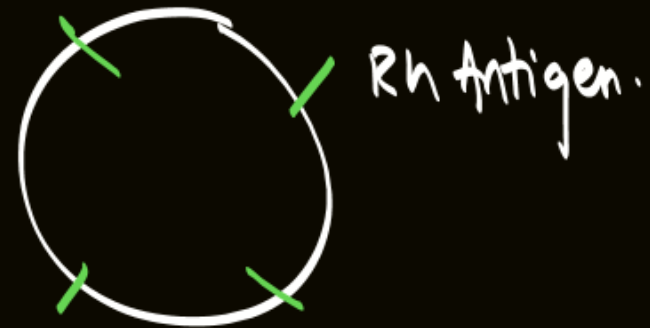
1. A +



2. AB +



3. B -



4. O +

Table of Transfusion

O- = 

Blood group

Donor

A+

A+, A-, O+, O- ✓

A-

A-, O-

B+

B+, B-, O+, O-

B-

B-, O-

AB+

AB+, AB-, A+, A-, B+, B-, O+, O- — Universal Donor

Universal Acceptor

AB-

AB-, A-, B-, O-

O+

O+, O-

O-

O-

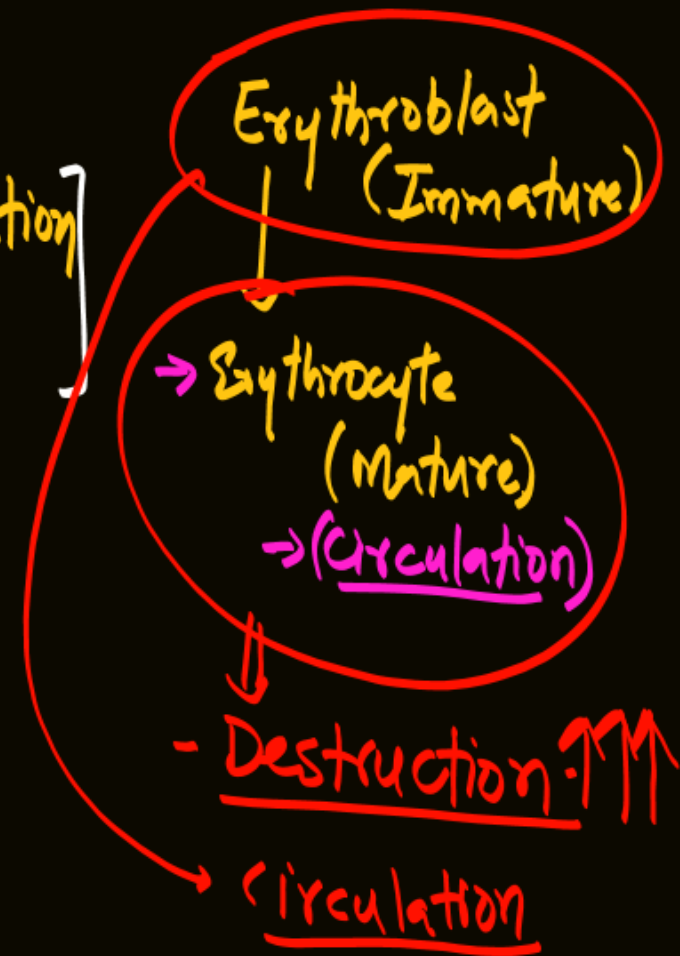
Rh incompatibility

- It arises when Rh $\oplus$ Blood is given to Rh $\ominus$ person.
- Rh $\ominus$ person will produce Antibodies against Rh $\oplus$ Blood. i.e. Anti Rh Antibodies (Rh Antibodies)

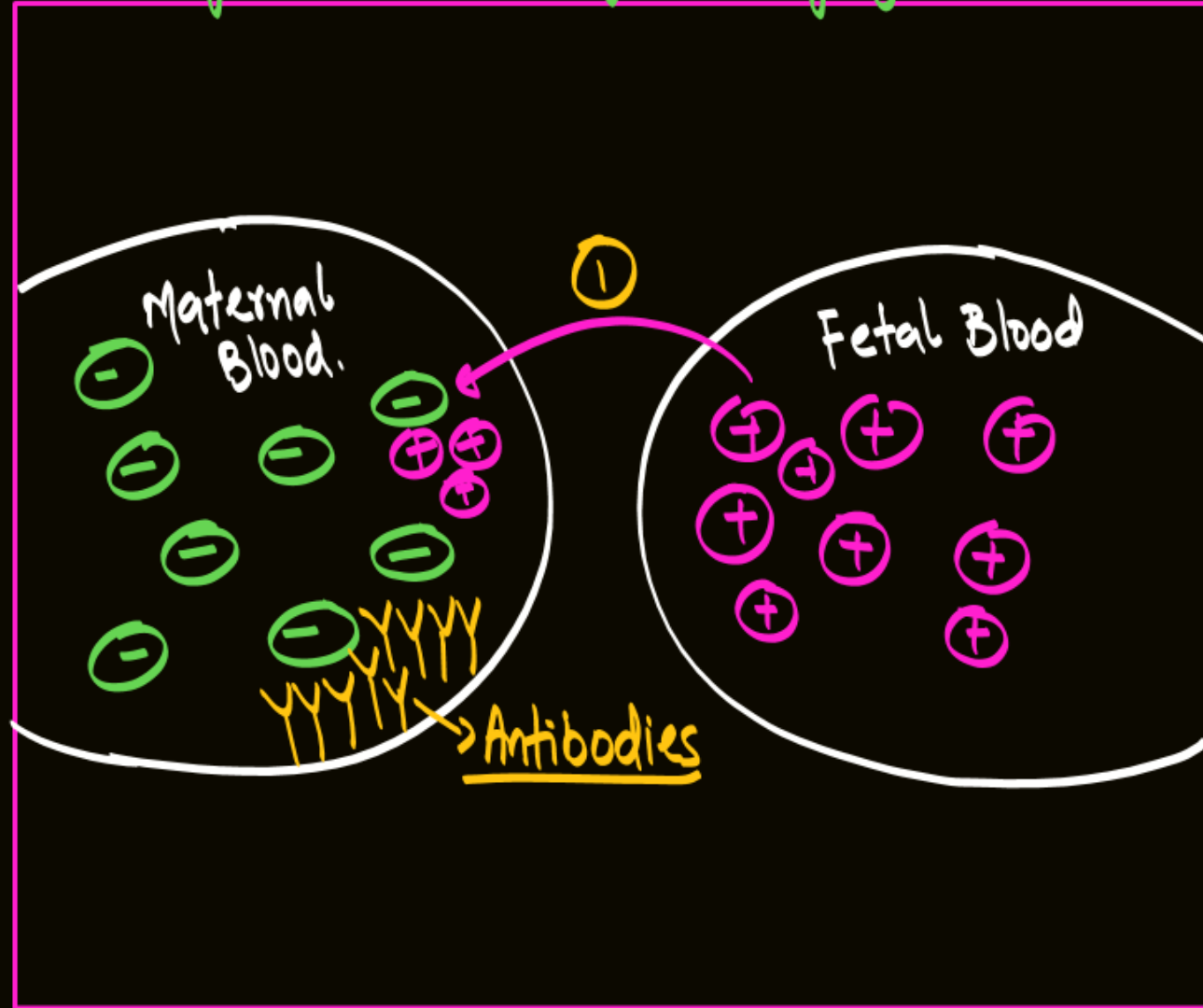
e.g. Erythroblastosis Foetalis

Definition - Literally it means Rise of Erythroblasts in Fetal circulation due to Damage to Erythrocytes.
→ [Foetus = 8 Week - End pregnancy in Mother's Uterus].

Reason - Here Mother is Rh(-) and Fetus is Rh(+), and if Blood of Fetus enters Mother's Side, so Mother will start producing Rh Antibodies, These Antibodies will cross towards Fetal side and destroys Fetal Erythrocytes.

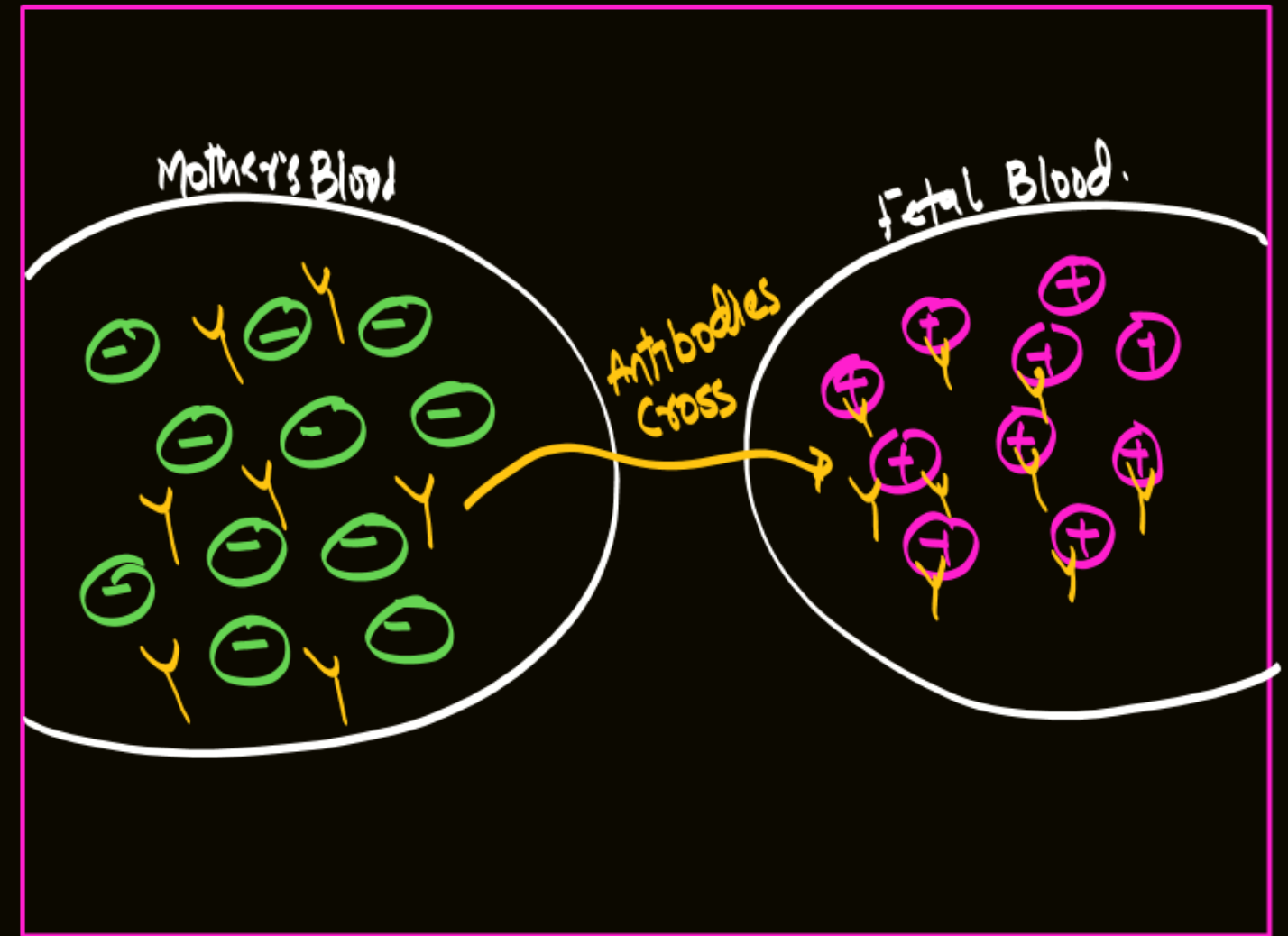


→ Blood gets Mixed during delivery of First child (+).



- Here Mother's Immune cells produce Antibodies if Fetal (+) Blood enters Mother side.

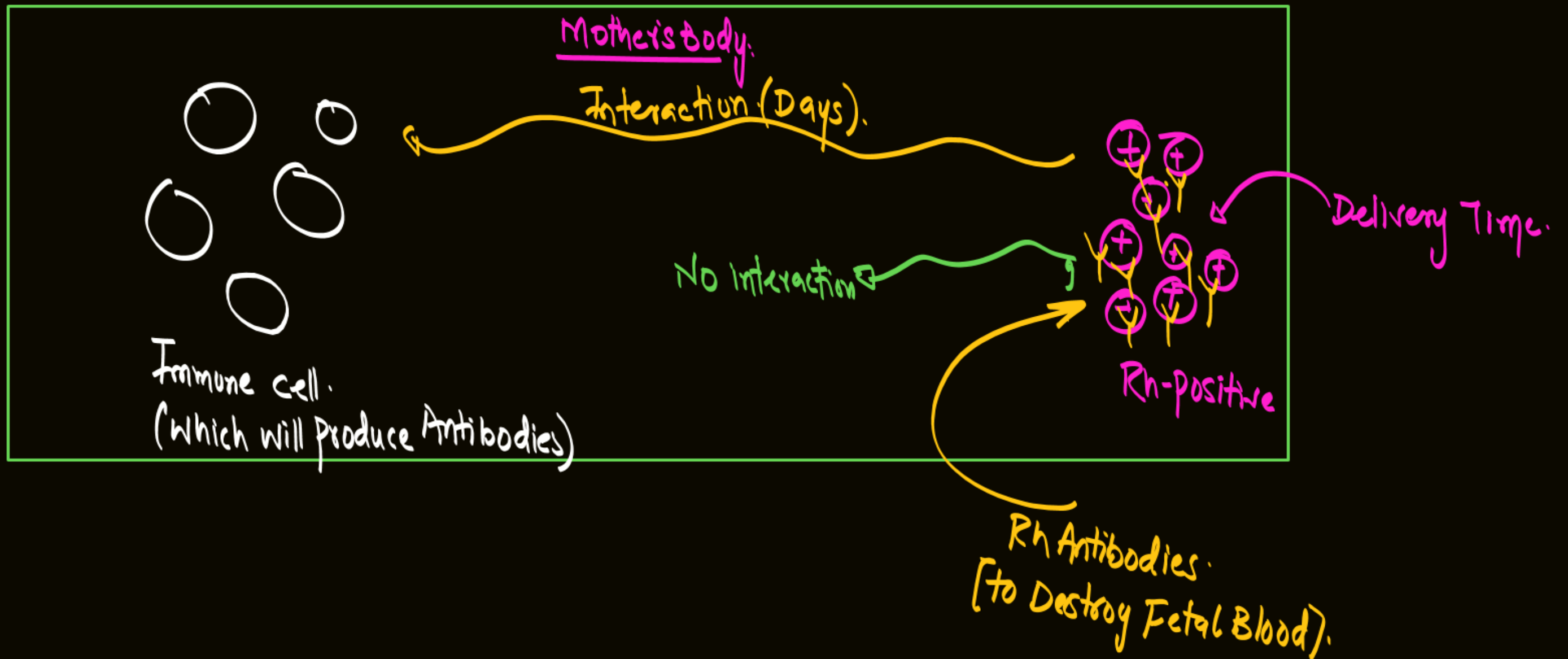
→ In Second (next) Pregnancy, if child is Again (+)



- In the next pregnancy Antibodies May cross, and destroy Fetal Erythrocytes, Causes = Hemolytic Diseases of Newborn [HDN].
- Newborn has - Anemia, Jaundice

Prevention.

- Here Anti Rh Antibodies are given ^{↑ To Mother} within 72 hours after delivery of First pregnancy / child.



Clotting

- Also k/as Coagulation

- It is a Step in Hemostasis [Stoppage of Bleeding]

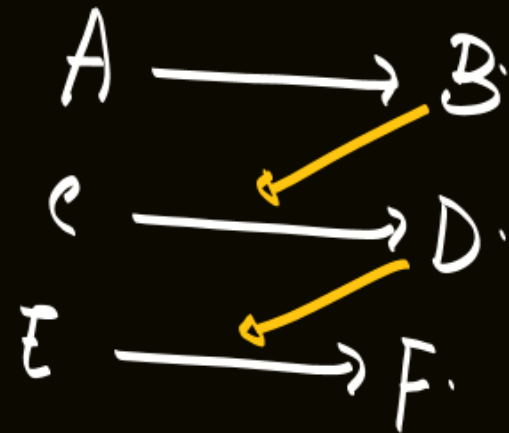
→ Coagulation / Clotting is a process in which the liquid form of Blood is converted into Solid (Non-Flowable), so that Bleeding Stops at the site of Injury.

→ It requires 13 Clotting Factors [Factor VII is Hypothetical].

→ Clotting is explained by Cascade Theory.

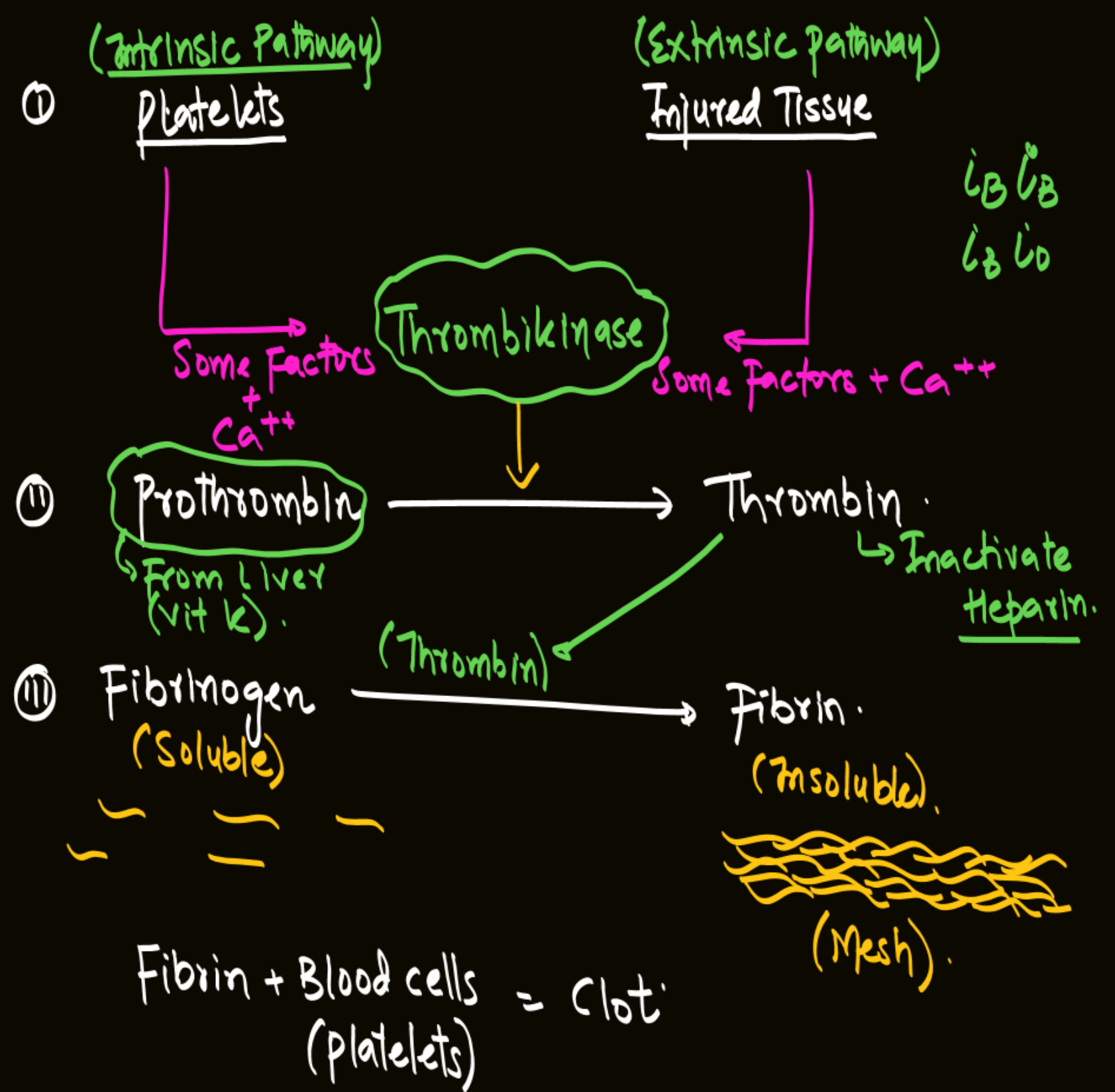
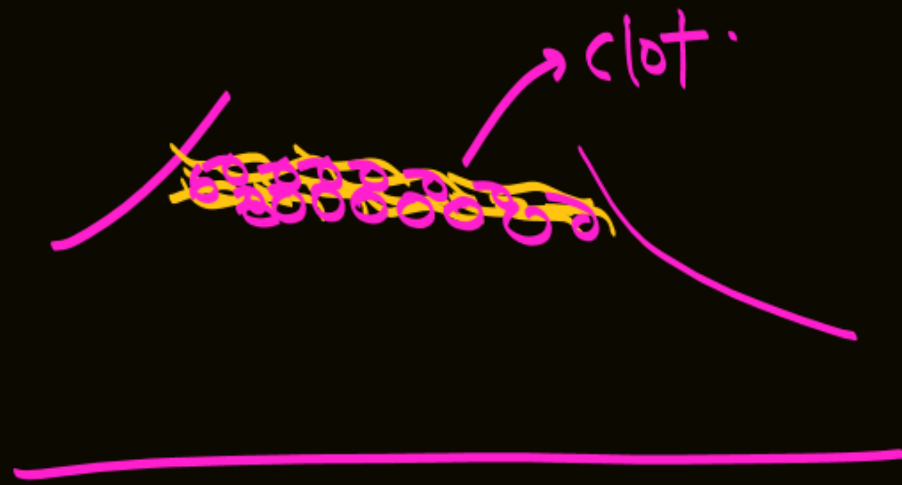
→ Multistep process.

→ Each product of Reaction drives next Reaction.

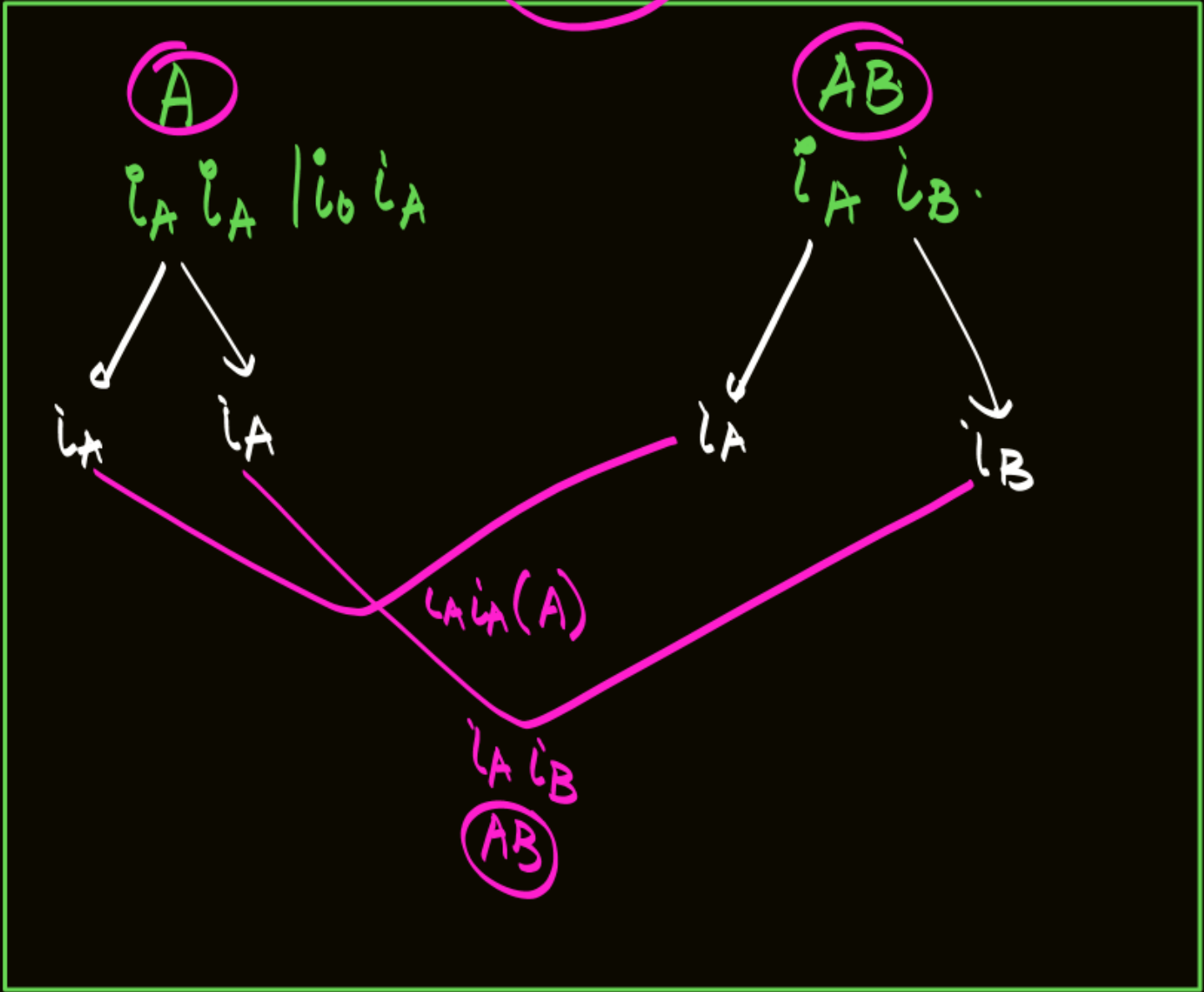


Clotting occurs in three steps.

- ① Formation of Thrombokinase.
- ② Formation of Thrombin.
- ③ Formation of Fibrin.



Doubt



Clotting Factors

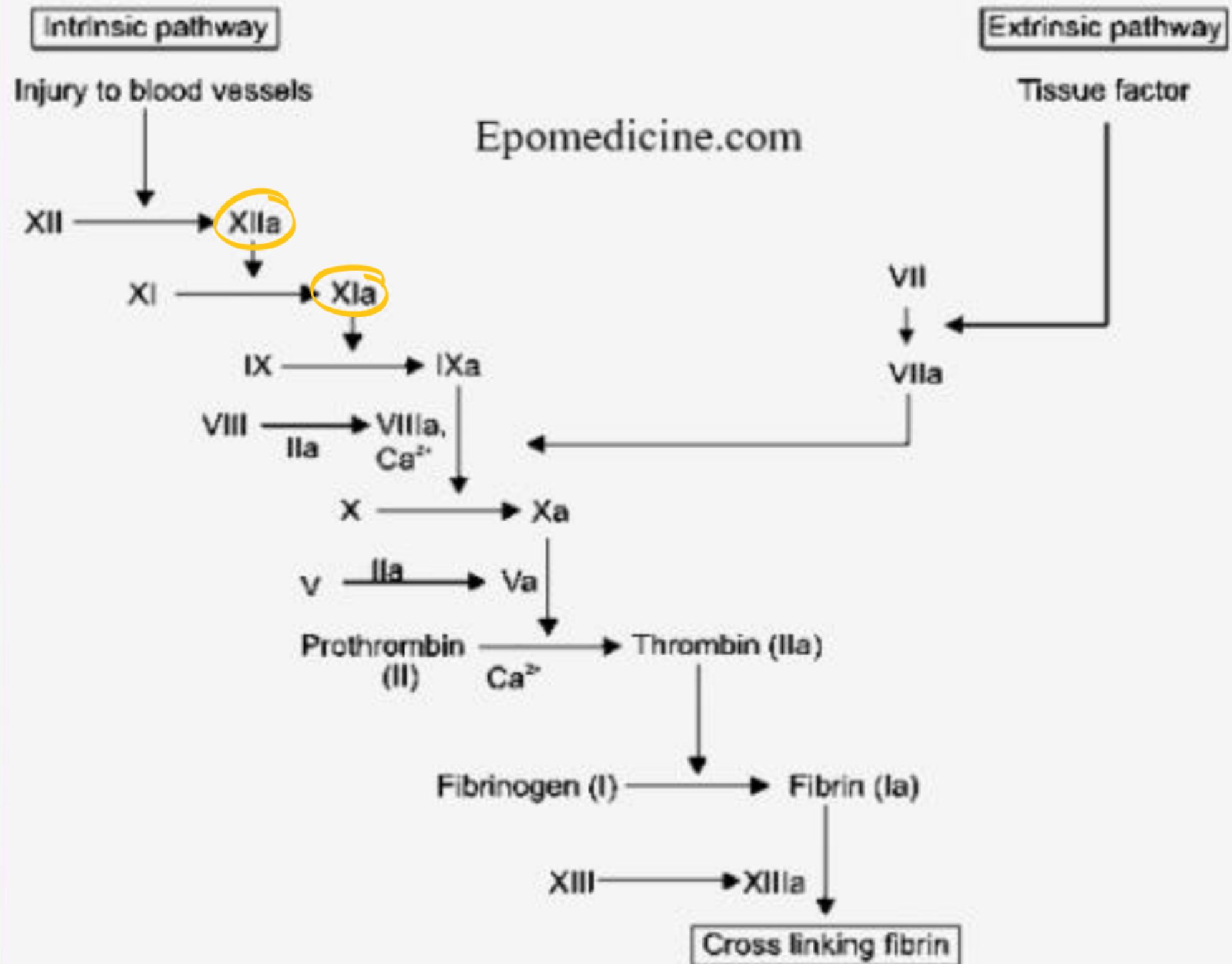
Foolish People Try Climbing Long
Slopes After Christmas, Some
People Have Fallen

Factor I	F	Fibrinogen
Factor II	P	Prothrombin
Factor III	T	Thromboplastin
Factor IV	C	Calcium
Factor V	L	Labile factor
Factor VII	S	Stable factor
Factor VIII	A	Antihemophilic factor A
Factor IX	C	Christmas factor
Factor X	S	Stuart Power factor
Factor XI	P	Plasma thromboplastin antecedent
Factor XII	H	Hagman factor
Factor XIII	F	Fibrin stabilising factor



VI





Thank
You